Q3

Group

```r
# Load the dataset. If 'mammals.csv' is not in your working directory,
# the code will download it from a public repository.
getwd()
```

```
## [1] "C:/Users/winni/Documents/MIDS/mids203HW/Project2"
```

```r
file_path <- "/Users/winni/Documents/MIDS/mids203/data/mammals.csv"

  df <- read.csv("C:\\Users\\winni\\Documents\\MIDS\\mids203\\data\\mammals.csv")
  print("Loaded data from URL.")
```

```
## [1] "Loaded data from URL."
```

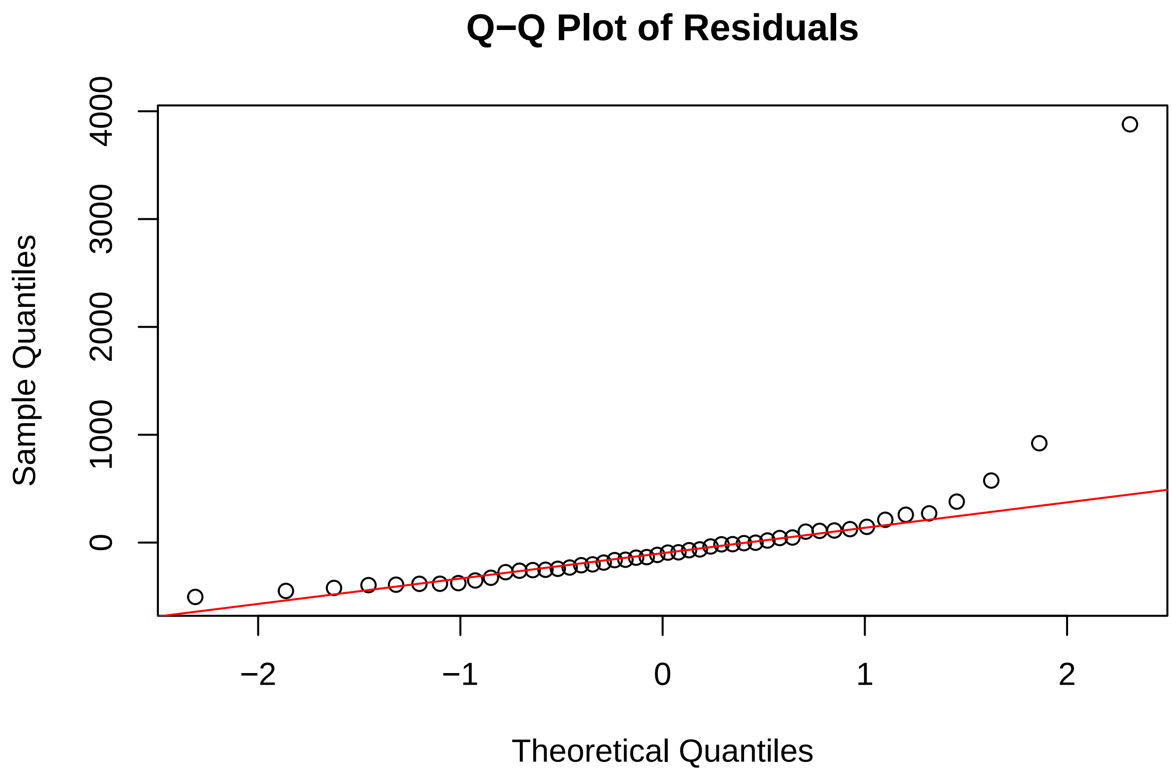
```r
# Prepare the data by removing rows with missing values
df_clean <- df %>%
  select(brain_wt, dreaming, non_dreaming) %>%
  drop_na()

# Estimate the linear model
model <- lm(brain_wt ~ dreaming + non_dreaming, data = df_clean)

# Print the model summary
summary(model)
```

```
##
## Call:
## lm(formula = brain_wt ~ dreaming + non_dreaming, data = df_clean)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -504.0 -256.9 -103.7   60.7 3878.6
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    776.00     243.05   3.193  0.00257 **
## dreaming        68.28      76.23   0.896  0.37513
## non_dreaming   -83.10      30.04  -2.767  0.00819 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 647.5 on 45 degrees of freedom
## Multiple R-squared:  0.1515, Adjusted R-squared:  0.1137
## F-statistic: 4.016 on 2 and 45 DF,  p-value: 0.02484
```

title: "Q3.1" output: pdf_document —

Q3.1 Independent and Identically Distributed (IID)

This dataset is likely IID. The data for each mammal has been collected from the same study and taken as an average of the population.Visually, they also appear to be mostly identically distributed along our plot without clustering. In addition, the individual data collected from these mammals appear independent of one another without dependencies.



— title: “Q3.2” output: pdf_document —

Q3.2 No Perfect Collinearity

The No Perfect Collinearity assumption requires that independent variables are not perfectly linearly related. We can check this with a correlation matrix and Variance Inflation Factors.

```
## [1] "Correlation Matrix:"
```

```
##               dreaming non_dreaming
## dreaming     1.0000000    0.5142539
## non_dreaming 0.5142539    1.0000000
```

```
## [1] "VIF Scores:"
```

```
##     dreaming non_dreaming
##      1.35954      1.35954
```

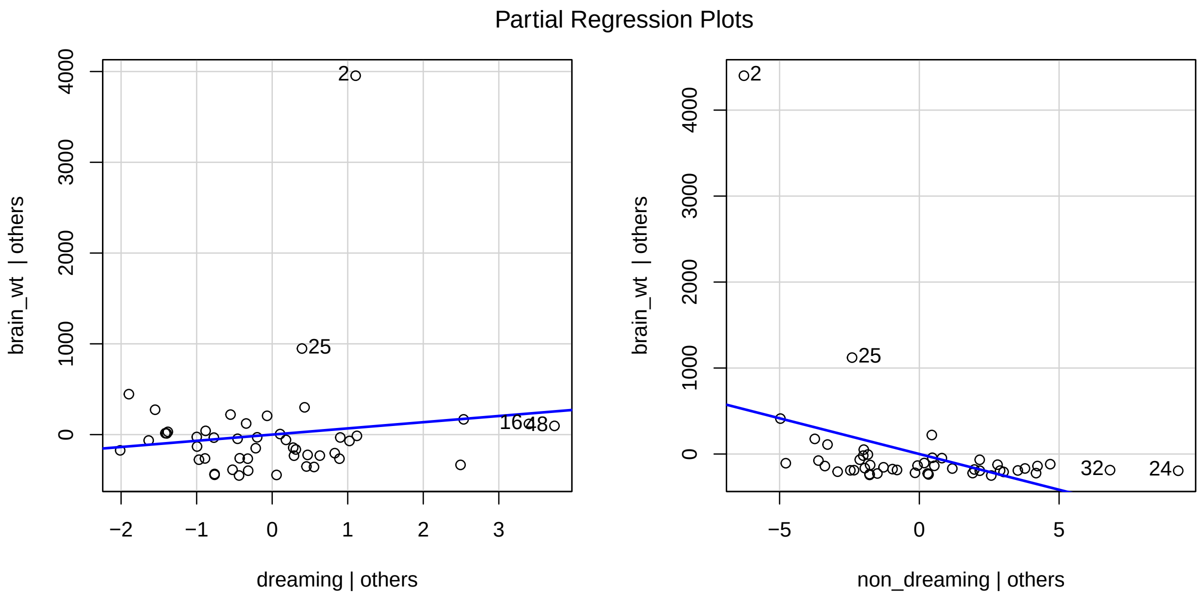
The correlation between dreaming and non_dreaming is -0.53, which is moderate. The VIF scores for both predictors are 1.39, well below the common threshold of 5 or 10. Though upon first glance, these variables may seem like perfect collinearity, this does not allign with the data.

This model has no perfect collinearity due to this data.

— title: “Q3.3” output: pdf_document —

Q3.3 Linear Conditional Expectation

The relationship between the predictors and the outcome should be linear. We can check this using partial regression plots.



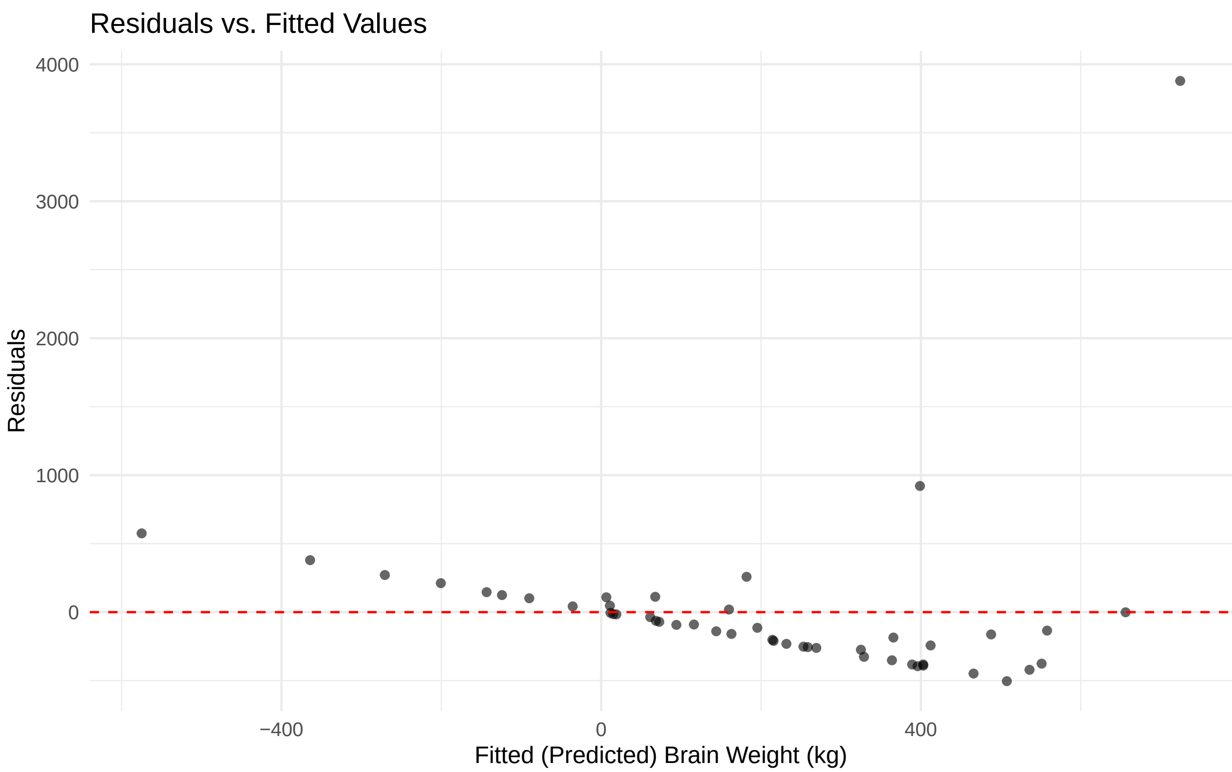
The partial regression plot for non_dreaming shows a reasonable positive linear trend. However, the plot for dreaming reveals a much weaker relationship, with the trend being heavily influenced by a number of outliers.

The assumption seems unlikely. The relationship with non_dreaming appears roughly linear, but the relationship with dreaming is weak and non-linear with it’s litany of outliers.

— title: "Q3.4" output: pdf_document —

Q3.4 Constant Error Variance (Homoskedasticity)

The variance of the model's errors should be constant across all levels of the predicted values. We check this by plotting the model's residuals against its fitted (predicted) values.



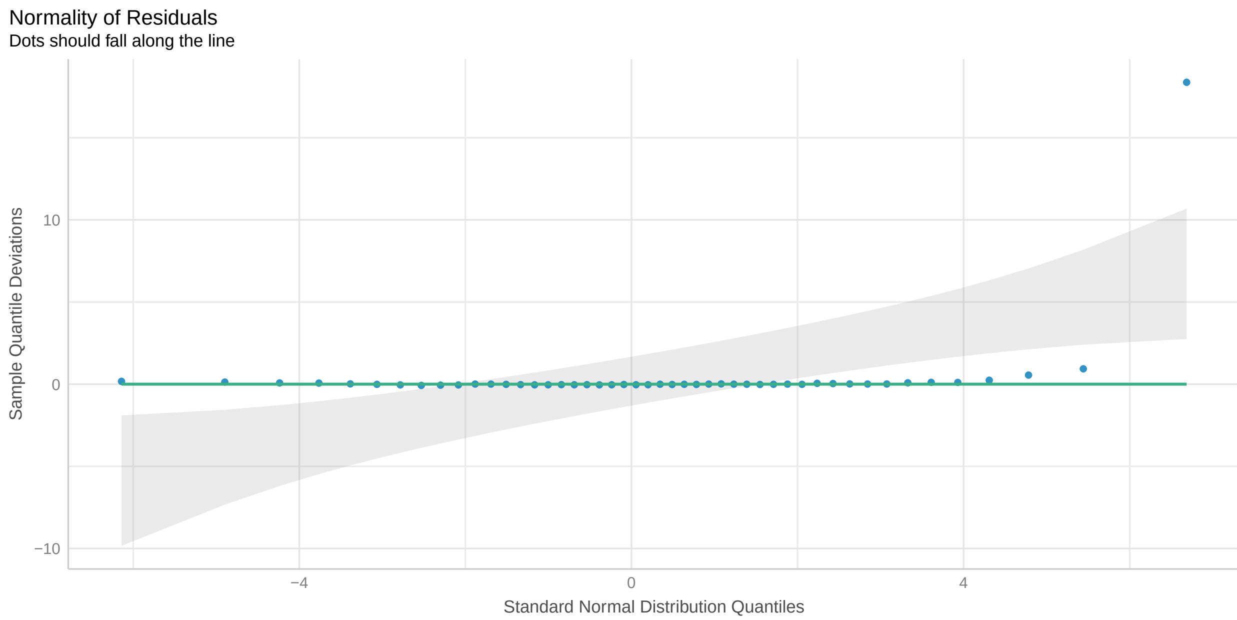
The plot clearly displays a funnel or cone shape. The vertical spread of the residuals increases as the fitted brain weight increases. This is a sign of heteroskedasticity.

The assumption of homoskedasticity is violated here.

— title: "Q3.5" output: pdf_document —

Q3.5 Normal Distribution of Errors

The model's errors should follow a normal distribution. We evaluate this with a Q-Q plot and a histogram.



The histogram of the residuals is heavily right-skewed and not bell-shaped. On the Q-Q plot, the points deviate significantly from the theoretical diagonal line, especially in the right tail. This indicates the residuals have heavier tails than a normal distribution.

The assumption of normally distributed errors is clearly violated, likely due to the presence of large-brained outliers.